Basics of Functions

Q1. Functions or not? For each of the following, decide whether it is a function or not and give a short explanation:

(a) $f(x) = x^2 + x - 3$

(b) $f\colon \mathbb{R} \longrightarrow \mathbb{R}$ given by $f(x) = \dfrac{\sin(x)}{x-1}$

(c) $f\colon (-\pi, \pi) \longrightarrow \mathbb{R}$ given by $f(x) = \tan(x/2)$

Q2. Image of a Set under a Function. Find the image of the interval $(0, \pi]$ under each of the following functions:

(a) $f\colon \mathbb{R} \longrightarrow \mathbb{R}$ given by $f(x) = 3 - 2x$

(b) $g\colon \mathbb{R} \longrightarrow \mathbb{R}$ given by $g(x) = e^x$

(c) $h\colon \mathbb{R} \longrightarrow \mathbb{R}$ given by $h(x) = \sin(x)$

Q3. Injective Functions. For each of the following functions, find the largest intervals on which it is injective:

(a) $f\colon \mathbb{R} \longrightarrow \mathbb{R}$ given by $f(x) = 3 - 2x$

(b) $g\colon \mathbb{R} \longrightarrow \mathbb{R}$ given by $g(x) = e^{|x|}$

(c) $h\colon \mathbb{R}\backslash\{k\pi/2 \mid k \in \mathbb{Z}\} \longrightarrow \mathbb{R}$ given by $h(x) = \operatorname{cosec}(2x)$

Q4. Counting Functions. Suppose $n \in \mathbb{N}$. How many functions $f\colon \{a, b, c\} \longrightarrow \{1, 2, \ldots, n\}$ are there?

Q5. Counting Injective Functions. Suppose $n \in \mathbb{N}$. How many injective functions $f\colon \{a, b, c\} \longrightarrow \{1, 2, \ldots, n\}$ are there:

(a) if $n < 3$?

(b) if $n \geq 3$?

Q6. Counting Bijective Functions. Suppose $n \in \mathbb{N}$. How many bijective functions $f\colon \{1, 2, \ldots, n\} \longrightarrow \{a, b, c\}$ are there:

(a) if $n < 3$?

(b) if $n = 3$?

(c) if $n > 3$?

Q7. Inverse functions. For each of the following functions, either find the inverse or explain why the inverse does not exist:

(a) $f\colon (-\infty, 1] \longrightarrow [0, \infty)$ given by $f(x) = x^2 - 3x + 2$

(b) $f\colon (0, \infty) \longrightarrow [0, \infty)$ given by $f(x) = |\log(x)|$

(c) $f\colon \mathbb{R} \longrightarrow (0, \infty)$ given by $f(x) = 2e^x + 1$

Inverse Trigonometric Functions

Q8. Inverse Trigonometric. Evaluate each of the following in radians:

(a) $\arcsin\left(\frac{1}{2}\right)$ (b) $\arccos\left(\frac{\sqrt{3}}{2}\right)$ (c) $\arctan\left(-\sqrt{3}\right)$

Q9. Simplification. Simplify the following:

(a) $\sin\left(\arcsin\left(\frac{1}{2}\right)\right)$

(b) $\tan\left(\arcsin\left(\frac{1}{\sqrt{2}}\right)\right)$

(c) $\arccos\left(\sin\left(\frac{5\pi}{4}\right)\right)$

(d) $\cos\left(\arcsin\left(\frac{1}{2}\right)\right)$

(e) $\arctan\left(2\cos\left(\frac{5\pi}{6}\right)\right)$

(f) $\arccos\left(\cos\left(\frac{11\pi}{6}\right)\right)$

Q10. Graphing Inverse Trigonometric Functions. On the same graph sketch

(a) $\arccos(x)$

(b) $\arccos(x/3)$

(c) $\arccos(x/3 - 1/3)$

(d) $2 - \arccos(x/3 - 1/3)$

Domain and Range

Q11. Step by Step. For each of part (a) and (b)

(i) Find the implied domain and range of f.
(ii) Find the implied domain and range of g.
(iii) Find $\text{dom}\,(f) \cap \text{range}\,(g)$.
(iv) Use your answers to (i), (ii) and (iii) to find the implied domain and range of $f \circ g$.

(a) $f(x) = \sqrt{x}$; $g(x) = 4 - x^2$; $(f \circ g)(x) = \sqrt{4 - x^2}$.

(b) $f(x) = \cot(x)$; $g(x) = \arcsin(x)$; $(f \circ g)(x) = \cot(\arcsin(x))$.

Q12. Standard. State the (i) implied domain and (ii) range, of each of the following:

(a) $y = \dfrac{1}{1-x}$

(b) $y = \log(1 + x)$

(c) $y = \sqrt{4 - x^2}$

(d) $y = \dfrac{1}{\sqrt{4 - x^2}}$

(e) $y = \sqrt{2\sin(x)}$

(f) $y = 2\log\left(\cos(x)\right)$

Q13. Inverse Trigonometric. State the (i) implied domain and (ii) range, of each of the following:

(a) $y = \arcsin(1 - x)$

(b) $y = \arccos(2x + 3)$

(c) $y = \arctan(4 - x)$

(d) $y = \arccos(\sin(2x))$

(e) $y = \tan(2\arcsin(x))$

(f) $y = \arccos(\sqrt{3}\tan(x))$

Q14. An incremental example. State the (i) implied domain and (ii) range, of each of the following:

(a) $y = |x|$

(b) $y = 2|x|$

(c) $y = 2|x| - 1$

(d) $y = \arcsin(2|x| - 1)$

Q15. Trigonometric. Express the following as algebraic functions of x:

(a) $f(x) = \cos(\arccos(x))$ (b) $g(x) = \sin(\arccos(x))$

State the implied domain and range of each function.

Answers

Topic 3: Functions

A1. Functions or not?

(a) not a function: no domain and codomain given; (b) not a function: undefined at $x = 1$; (c) function.

A2. Image of a Set under a Function. (a) $[3 - 2\pi, 3)$; (b) $(1, e^{\pi}]$; (c) $[0, 1]$.

A3. Injective Functions.

(a) $\mathbb{R}$; (b) $(-\infty, 0]$ and $[0, \infty)$; (c) $\ldots, (-\pi/2, -\pi/4], [-\pi/4, 0), (0, \pi/4], [\pi/4, \pi/2), (\pi/2, 3\pi/4], \ldots$.

A4. Counting Functions. n^3.

A5. Counting Injective Functions. (a) None. (b) $n(n-1)(n-2)$.

A6. Counting Bijective Functions. (a) None. (b) $6 = 1 \cdot 2 \cdot 3 = 3!$. (c) None.

A7. Inverse functions.

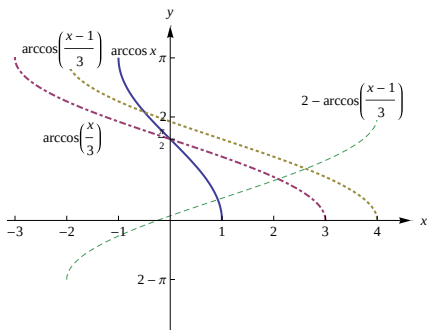
(a) $g(x) = \dfrac{3 - \sqrt{1 + 4x}}{2}$. (b) No inverse because f is not injective: $f(e) = f(e^{-1}) = 1$.

(c) No inverse because f is not surjective: the range of f is $(1, \infty) \subsetneq (0, \infty)$.

A8. Inverse Trigonometric. (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{6}$ (c) $-\frac{\pi}{3}$

A9. Simplification. (a) $\frac{1}{2}$ (b) 1 (c) $\frac{3\pi}{4}$ (d) $\frac{\sqrt{3}}{2}$ (e) $-\frac{\pi}{3}$ (f) $\frac{\pi}{6}$

A10. Graphing Inverse Trigonometric Functions.



A11. Step by Step. Let $\mathrm{dom}(f)$ denote the domain of f and $\mathrm{range}(f)$ denote the range of f.

(a)

(i) $\mathrm{dom}(f) = [0, \infty)$; $\mathrm{range}(f) = [0, \infty)$ (ii) $\mathrm{dom}(g) = \mathbb{R}$; $\mathrm{range}(g) = (-\infty, 4]$

(iii) $[0, \infty) \cap (-\infty, 4] = [0, 4]$ (iv) $\mathrm{dom}(f \circ g) = [-2, 2]$; $\mathrm{range}(f \circ g) = [0, 2]$

(b)

(i)$\mathrm{dom}(f) = \mathbb{R}\backslash\{k\pi : k \in \mathbb{Z}\} = \{(k\pi, \pi + k\pi) : k \in \mathbb{Z}\}$; $\mathrm{range}(f) = \mathbb{R}$
(ii) $\mathrm{dom}(g) = [-1, 1]$; $\mathrm{range}(g) = [-\pi/2, \pi/2]$
(iii) $\{(k\pi, \pi + k\pi) : k \in \mathbb{Z}\} \cap [-\pi/2, \pi/2] = [-\pi/2, 0) \cup (0, \pi/2]$
(iv) $\mathrm{dom}(f \circ g) = [-1, 0) \cup (0, 1]$; $\mathrm{range}(f \circ g) = \mathbb{R}$

A12. Standard.

(a) (i) $\mathbb{R}\backslash\{1\}$ (ii) $\mathbb{R}\backslash\{0\}$ (b) (i) $(-1, \infty)$ (ii) $\mathbb{R}$ (c) (i) $[-2, 2]$ (ii) $[0, 2]$

(d) (i) $(-2, 2)$ (ii) $[\frac{1}{2}, \infty)$ (e) (i) $[2k\pi, 2k\pi + \pi]$, $k \in \mathbb{Z}$ (ii) $[0, \sqrt{2}]$

(f) (i) $(-\frac{\pi}{2} + 2k\pi, \frac{\pi}{2} + 2k\pi)$, $k \in \mathbb{Z}$ (ii) $(-\infty, 0]$

A13. Inverse Trigonometric.

(a) (i) $[0, 2]$ (ii) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (b) (i) $[-2, -1]$ (ii) $[0, \pi]$ (c) (i) $\mathbb{R}$ (ii) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(d) (i) $\mathbb{R}$ (ii) $[0, \pi]$ (e) (i) $[-1, 1]\backslash\{\pm\frac{1}{\sqrt{2}}\}$ (ii) $\mathbb{R}$

(f) (i) $\left[-\frac{\pi}{6} + k\pi, \frac{\pi}{6} + k\pi\right]$, $k \in \mathbb{Z}$ (ii) $[0, \pi]$

A14. An incremental example.

(a) (i) $\mathbb{R}$ (ii) $[0, \infty)$ (b) (i) $\mathbb{R}$ (ii) $[0, \infty)$ (c) (i) $\mathbb{R}$ (ii) $[-1, \infty)$ (d) (i) $[-1, 1]$ (ii) $[-\pi/2, \pi/2]$

A15. Trigonometric.

(a) $f(x) = x$; domain is $[-1, 1]$; range is $[-1, 1]$ (b) $g(x) = \sqrt{1 - x^2}$; domain is $[-1, 1]$; range is $[0, 1]$

Useful Formulae

Pythagorean identity

$$\cos^2(x) + \sin^2(x) = 1$$

Compound angle formulae

$$\begin{aligned}
\sin(x+y) &= \sin(x)\cos(y) + \cos(x)\sin(y) \\
\cos(x+y) &= \cos(x)\cos(y) - \sin(x)\sin(y) \\
\tan(x+y) &= \frac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}
\end{aligned}$$

Derivatives of inverse trigonometric functions

$$\begin{aligned}
\arcsin'(x) &= \frac{1}{\sqrt{1-x^2}} \\
\arccos'(x) &= -\frac{1}{\sqrt{1-x^2}} \\
\arctan'(x) &= \frac{1}{1+x^2}
\end{aligned}$$

Antiderivatives from inverse trigonometric functions

$$\begin{aligned}
\int \frac{1}{\sqrt{s^2-x^2}}\,dx &= \arcsin\left(\frac{x}{s}\right) + C \\
\int \frac{-1}{\sqrt{s^2-x^2}}\,dx &= \arccos\left(\frac{x}{s}\right) + C \\
\int \frac{1}{s^2+x^2}\,dx &= \frac{1}{s}\arctan\left(\frac{x}{s}\right) + C
\end{aligned}$$

where s is a *positive* constant, and C is an arbitrary constant of integration.

Complex exponential formulae

$$\begin{aligned}
e^{i\theta} &= \cos(\theta) + i\sin(\theta) \\
\cos(\theta) &= \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) \\
\sin(\theta) &= \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)
\end{aligned}$$

Vector projections

- The *vector projection* of $\mathbf{v}$ onto $\mathbf{u}$ is $\mathbf{v}_{\parallel} = (\hat{\mathbf{u}} \cdot \mathbf{v})\hat{\mathbf{u}} = k\mathbf{u}$, where $k \in \mathbb{R}$ is the unique solution of $\mathbf{u}\cdot(\mathbf{v}-k\mathbf{u}) = 0$.
- The *vector component of* $\mathbf{v}$ *perpendicular to* $\mathbf{u}$ is $\mathbf{v}_{\perp} = \mathbf{v} - \mathbf{v}_{\parallel}$.

Complex roots

The n-th roots of $w = se^{i\phi}$ are $s^{\frac{1}{n}}\, e^{i(\frac{1}{n}(\phi+2k\pi))}$ for $k = 0, 1, \ldots, n-1$.

Changes in speed

Provided $\mathbf{r}'(t) \neq \mathbf{0}$, the speed function $\|\mathbf{r}'(t)\|$ is decreasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) < 0$ and increasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) > 0$.